Ishrath Ahamed

ishrathahamed0@gmail.com | +94 77 147 6789
linkedin.com/in/ishrathahamed | github.com/ishrath99 | ishrath99.github.io

Experience

Software Engineer Jun. 2024 – Present
Cloud Solutions International (Pvt) Ltd *Sri Lanka*

- Designed and developed scalable, interoperable EHR microservices using Python (Django) and Java (Spring Boot), including IVF workflows
- Implemented agentic orchestration workflows with MCP servers, RAG, and Neo4j knowledge graphs
- Built a production-grade, real-time voice AI agent for an Android app over a low-latency WebRTC streaming pipeline
- Implemented a Cerbos policy manager service in Go for resource permissions and improved security service performance
- Established API Gateway policy baselines (CORS, rate limits, security headers, public endpoint exception lists)
- Built internal platforms to track microservice release dependencies and automated CI/CD pipelines using Jenkins, Liquibase, and ArgoCD
- Led QA environment setup and managed Kubernetes-based deployments and containerized infrastructure
- Tech Stack: Go, Python, Django, Java, Spring Boot, Cerbos, Angular, Flutter, WebRTC, Kubernetes, Docker, Jenkins, Liquibase, ArgoCD, Neo4j

Visiting Research Fellow Jan. 2023 – Jun. 2023
Singapore University of Technology and Design *Singapore*

- Developed novel computer vision algorithms for real-world surveillance problems using deep learning
- Designed and trained video-based activity recognition models
- Collaborated with researchers to translate findings into deployable prototypes
- Tech Stack: Python, TensorFlow, Keras, OpenCV

Software Engineer Intern Jan. 2021 – Jun. 2021
Kairos Sensing *Sri Lanka*

- Developed Sri Lanka's first IMU sensor-based full-body 3D motion capture system
- Implemented core motion processing algorithms in Java and C++
- Integrated sensor data pipelines with hardware and firmware teams
- Tech Stack: Java, C++

Education

University of Moratuwa Sri Lanka
B.Sc.(Hons) Biomedical Engineering *Jan. 2020 – Jun. 2024*

- Cumulative GPA: 4.04/4.20 | First Class (Honours)

Royal College, Colombo - 07 Sri Lanka
G.C.E. Advanced Level Examination (Physical Science) *Jun. 2016 - Aug. 2018*

- Z-Score: 2.4 (3As) | Ranked 83rd in the country

Publications

Real-Time AI-Driven People Tracking and Counting Using Overhead Cameras | IEEE-Explore 2024
Presented in TENCON 2024 - 2024 IEEE Region 10 Conference (TENCON) - Singapore

Selected Projects

Julia – Voice AI Nurse Assistant 2026–Present

Real-time, voice-first AI assistant for clinical nurses (Android + WebRTC)

- Built a real-time voice assistant for nurses to query EHR data, guidelines, tasks, and medications by voice
- Engineered a low-latency WebRTC voice pipeline (Pipecat) streaming STT, LLM, and TTS
- Designed a privacy-by-construction architecture keeping patient identity server-side, never exposed to the LLM
- Implemented a plugin-based provider/adapter system for EHR, task, and medication integrations with safe fallbacks
- Developed a Flutter Android client for hands-free nurse interaction over WebRTC
- Tech Stack: Python, FastAPI, Pipecat, WebRTC, Neo4j, MCP, Flutter, Dart

IVF Module for EHR Systems 2025–2026

Production-grade IVF platform integrated with enterprise EHR systems

- Architected IVF-specific database schemas for treatment cycles, lab workflows, and longitudinal patient records
- Implemented Spring Boot backend exposing IVF services as a standalone, EHR-agnostic module
- Developed Angular frontend to extend EHR UI with IVF workflows, dashboards, and clinical views
- Integrated the module with core EHR via Kafka event streaming and REST APIs for real-time interoperability
- Ensured alignment between IVF workflows and EHR entities such as patient profiles, encounters, and records
- Tech Stack: Java, Spring Boot, Angular, TypeScript, HTML, Kafka, REST APIs

TwinDoctor 2025

Agentic workflow platform with multi-agent orchestration and graph-based intelligence

- Implemented agentic workflows with coordinated multi-agent orchestration using MCP servers
- Modeled complex entity relationships in Neo4j for context-aware data retrieval
- Built RAG pipelines to enhance agent reasoning with healthcare knowledge bases
- Implemented authentication and access control for secure agent interactions
- Contributed bug fixes to the open-source Agno agent framework
- Tech Stack: Python, Neo4j, MCP, RAG, Agno

Release Automation Platform 2024–2025

Enterprise platform for microservice releases and dependency orchestration

- Designed centralized data model to manage dependencies across microservices
- Built Django web app to manage release metadata
- Developed Jenkins pipelines to automate ingestion and validation of dependency data
- Reduced release time from 1 hour to under 10 minutes by automating database migrations and workflows
- Executed Python-based ArgoCD workflows for zero-downtime releases
- Tech Stack: Python, Django, Jenkins, Liquibase, ArgoCD, JavaScript

Computational Microscope Design and 3D Image Reconstruction | github 2024

Undergraduate thesis in collaboration with Harvard University

- Investigated deep learning approaches for solving ill-posed inverse problems in 3D image reconstruction
- Designed and modeled optics-inspired computational imaging systems
- Tech Stack: Python, MATLAB, PyTorch

Edge-based Activity Recognition Systems 2023

Real-time computer vision solutions deployed on constrained devices

- Developed 3D CNN-based video classification models to detect illegal activities in residential environments
- Implemented face recognition pipelines using deep feature embeddings for secure access control systems
- Designed lightweight people-counting and object-tracking algorithms optimized for edge deployment
- Tech Stack: Python, TensorFlow, Keras, OpenCV

Walksense – 3D Motion Capture System | github 2021

Sri Lanka's first low-cost full-body 3D motion capture system

- Integrated wireless IMU sensors and developed motion analysis algorithms for full-body tracking
- Implemented real-time visualization and motion processing pipelines
- Tech Stack: Java, C++, jMonkeyEngine

Skills

Programming Languages: Python, Java, Go, SQL, TypeScript, Dart, MATLAB,
Frameworks & Libraries: Django, Spring Boot, FastAPI, Cerbos, Angular, React, Flutter, Pipecat, WebRTC, Liquibase
Databases: PostgreSQL, Oracle, MongoDB, Neo4j, Redis
DevOps & Tools: Kubernetes, ArgoCD, Jenkins, Docker, Git, GitHub
ML / AI / CV: PyTorch, TensorFlow, Keras, OpenCV

References

Raditha Dissanayake – Head of AI and Innovation, Cloud Solutions International, Sri Lanka.
Contact: +94 77 756 2242, Email: raditha.dissanayake@gmail.com

Tishan Harischandra – Associate Architect, Cloud Solutions International, Sri Lanka.
Contact: +94 71 804 5768, Email: chanakasandu@gmail.com